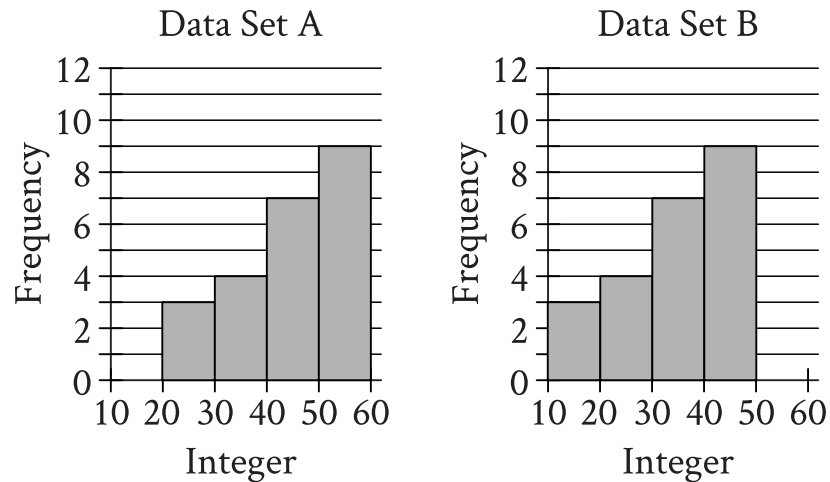


Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard

Question



Two data sets of **23** integers each are summarized in the histograms shown. For each of the histograms, the first interval represents the frequency of integers greater than or equal to **10**, but less than **20**. The second interval represents the frequency of integers greater than or equal to **20**, but less than **30**, and so on. What is the smallest possible difference between the mean of data set A and the mean of data set B?

Answer

- A. 0
- B. 1
- C. 10
- D. 23

Correct Answer: B

Rationale

Choice B is correct. The histograms shown have the same shape, but data set A contains values between **20** and **60** and data set B contains values between **10** and **50**. Thus, the mean of data set A is greater than the mean of data set B. Therefore, the smallest possible difference between the mean of data set A and the mean of data set B is the difference between the smallest possible mean of data set A and the greatest possible mean of data set B. In data set A, since there are **3** integers in the interval greater than or equal to **20** but less than **30**, **4** integers greater than or equal to **30** but less than **40**, **7** integers greater than or equal to **40** but less than **50**, and **9** integers greater than or equal to **50** but less than **60**, the smallest possible mean for data set A is $\frac{(3 \cdot 20) + (4 \cdot 30) + (7 \cdot 40) + (9 \cdot 50)}{23}$. In data set B, since there are **3** integers greater than or equal to **10** but less than **20**, **4** integers greater than or equal to **20** but less than **30**, **7** integers greater than or equal to **30** but less than **40**, and **9** integers greater than or equal to **40** but less than **50**, the largest possible mean for data set B is $\frac{(3 \cdot 19) + (4 \cdot 29) + (7 \cdot 39) + (9 \cdot 49)}{23}$. Therefore, the smallest possible difference between the mean of data set A and the mean of data set B is $\frac{(3 \cdot 20) + (4 \cdot 30) + (7 \cdot 40) + (9 \cdot 50)}{23} - \frac{(3 \cdot 19) + (4 \cdot 29) + (7 \cdot 39) + (9 \cdot 49)}{23}$, which is equivalent to $\frac{(3 \cdot 20) - (3 \cdot 19) + (4 \cdot 30) - (4 \cdot 29) + (7 \cdot 40) - (7 \cdot 39) + (9 \cdot 50) - (9 \cdot 49)}{23}$. This expression can be rewritten as $\frac{3(20 - 19) + 4(30 - 29) + 7(40 - 39) + 9(50 - 49)}{23}$, or $\frac{23}{23}$, which is equal to **1**. Therefore, the smallest possible difference between the mean of data set A and the mean of data set B is **1**.

Choice A is incorrect. This is the smallest possible difference between the ranges, not the means, of the data sets.

Choice C is incorrect. This is the difference between the greatest possible mean, not the smallest possible mean, of data set A and the greatest possible mean of data set B.

Choice D is incorrect. This is the smallest possible difference between the sum of the values in data set A and the sum of the values in data set B, not the smallest possible difference between the means.